

Mohit Kirtane

AI Engineer | Agentic AI | LLMs | RAG Systems | Backend Developer

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📍 Pune, Maharashtra

EXPERIENCE

Actin Technologies

July 2025 – Present

AI Engineer

Pune, India

- Engineered a production-grade **Enterprise AI Platform** integrating document intelligence, SQL chat, RBAC-based policy retrieval, and invoice extraction using **LLMs, RAG, LangChain, LangGraph, FastAPI, MongoDB, PostgreSQL**, and vector databases, enabling users to access enterprise knowledge through a unified conversational interface.
- Developed a **Vision AI Platform** supporting facial recognition, object detection, video understanding, conversational video search, and real-time analytics across recorded videos and live streams, significantly improving investigation and surveillance workflows.
- Designed and implemented an **AI-assisted Data Annotation Platform** leveraging computer vision models and LLMs to automate image and text annotation, accelerating dataset preparation and reducing manual labeling through human-in-the-loop validation.
- Fine-tuned and optimized an **facial recognition model** on custom datasets, improving recognition accuracy and deploying the model for reliable real-time inference in production environments.
- Built scalable FastAPI microservices, reusable AI pipelines, and LangGraph-based agentic workflows, enabling rapid deployment of multiple enterprise AI applications while integrating structured, unstructured, and vectorized data sources.

Atomic Loops

Jan 2024 – Apr 2024

Frontend Development Intern

Pune, India

- Developed responsive and user-friendly web applications using **React.js, JavaScript, Tailwind CSS, HTML**, and **CSS**, improving usability and frontend performance.
- Collaborated with backend developers to integrate REST APIs, optimize application workflows, and deliver scalable production-ready features within agile development cycles.

Elite Software

Nov 2022 – Mar 2023

Web Development Intern

Pune, India

- Developed full-stack web applications using **Python, Django, MySQL, HTML, CSS**, and **JavaScript**, delivering business-focused solutions for client requirements.
- Designed RESTful APIs, optimized database queries, and collaborated with cross-functional teams to enhance application performance and maintainability.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Generative & Agentic AI: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, AI Agents, Agentic Workflows, MCP, Function Calling

Machine Learning: Transformers, PyTorch, TensorFlow, Scikit-learn, Hugging Face, NLP, OpenCV

Databases: MongoDB, PostgreSQL, MySQL, FAISS, ChromaDB

Backend: FastAPI, Flask, REST APIs, gRPC, JWT Authentication

Developer Tools: Docker, Git, GitHub, Postman, ThunderClient, Azure DevOps, Ffmpeg

PROJECTS

Enterprise Knowledge Copilot | LangChain, LangGraph, FastAPI, MongoDB, PostgreSQL, FAISS, LLMs

- Engineered a unified enterprise AI platform supporting **4 core workflows**—document intelligence, database chat, RBAC-based policy retrieval, and invoice intelligence—using LLMs, RAG, LangChain, and LangGraph.
- Implemented **RBAC-aware retrieval** to enforce permission-controlled access to enterprise documents and policy knowledge, improving security across role-specific conversational workflows.
- Built natural-language-to-SQL capabilities enabling non-technical users to query structured databases conversationally, reducing dependency on manual SQL-based data retrieval.
- Designed LangGraph-based agentic workflows for query planning, retrieval, reasoning, and tool execution, improving flexibility and reliability across multi-step AI tasks.
- Integrated MongoDB, PostgreSQL, FAISS, and FastAPI into reusable AI services, enabling multiple enterprise workflows through a unified platform architecture.

Multimodal Video Intelligence Platform | Python, PyTorch, OpenCV, LangChain, Vector DB, gRPC, FFmpeg

- Built a multimodal video intelligence platform supporting **2 video contexts**—recorded footage and live streams—with natural-language search and conversational video analysis.
- Developed pipelines combining **object detection, tracking, object cropping, metadata indexing, and video description** to enable targeted retrieval of relevant visual events.
- Fine-tuned and integrated an **ArcFace facial recognition model** for identity-based search and automated person recognition across video streams.
- Combined visual embeddings, video descriptions, and LLM-powered question answering to enable contextual natural-language queries over video content.
- Designed high-performance inference pipelines using **PyTorch, gRPC, and FFmpeg** for scalable video processing and real-time analytics.

AI Data Annotation & Labeling Platform | Python, PyTorch, FastAPI, MongoDB, OpenCV, YOLO, LLMs

- Built an AI-assisted annotation platform supporting **2 data modalities**—image and text—to automate dataset preparation for machine learning workflows.
- Integrated **YOLO, computer vision models, and LLMs** to generate automated annotations, accelerating dataset preparation and reducing repetitive manual labeling.
- Developed human-in-the-loop validation workflows allowing users to review, correct, and approve AI-generated annotations, improving annotation consistency and quality.
- Designed FastAPI and MongoDB services for dataset management, annotation storage, processing, and integration with downstream model-training pipelines.

EDUCATION

Marathwada Mitra Mandal College of Engineering	2021 – 2025
Bachelor of Engineering (Computer Engineering) CGPA: 8.1/10	Pune, Maharashtra
Major Hemant Jakate School of Commerce & Science	2019 – 2021
Higher Secondary Certificate (HSC) Percentage: 99%	Nagpur, Maharashtra
The Aditya Birla Public School	2018
Secondary School Certificate (SSC) Percentage: 95%	Chandrapur, Maharashtra